

Bus a Dinner Table SOP

One episode buses one used placemat. The mat sits in the middle of the tabletop with a finished meal's tableware scattered loose across it. Nothing is stacked, nothing sits in a set position, the cutlery lies wherever it was put down, the napkin is crumpled, and scraps are left on the dishes.

Because the layout varies from episode to episode, **the clearing order is fixed by item type, not by position**: glass, then cutlery, then napkin, then dishes. That single rule is what keeps the data consistent across a scene that is different every time.

The table is cleared in two passes. **Pass 1** strips every item off the mat into the staging zone, shaking each dish over the waste bin on the way. **Pass 2** lays the bare placemat onto the tray as its base layer, then loads every staged item onto it in assigned zones.

The two receptacles sit on opposite sides of the tabletop. The **tray**, against the left edge, takes everything reusable: the placemat, dishes, cutlery, napkin, and glass. The **waste bin**, against the right edge, takes only the scraps shaken off the dishes; no object is ever put into it.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the tray on the left, the staging zone, the loaded placemat in the middle, and the waste bin on the right.
3. Both arms are at home with grippers open.
4. The tabletop is dry outside the mat.
5. Nothing obstructs the paths between the mat, the staging zone, the tray, and the waste bin.

Materials checklist

1. One placemat lies flat in the middle of the tabletop.
2. Scattered loose on the mat: a large plate, a small plate, a bowl, a fork, a knife, a soup spoon, and a cloth napkin. No dish is stacked on another dish.
3. The scatter is uneven, as a real finished meal leaves it: utensils resting on or beside dishes, the napkin crumpled rather than folded, items at no fixed angle.
4. All three dishes carry loose, dry food scraps.
5. Every item sits fully on the mat. Nothing overhangs onto the bare tabletop.
6. The tray is empty and lies flat. The waste bin is empty or partly filled, and it is not so full that scraps would sit above its rim.

7. The staging zone is clear.

Workspace layout

- **Tray:** on the tabletop against the left edge, within left arm reach.
- **Staging zone:** the clear tabletop between the mat and the tray, left of center. Both arms reach it. It stays empty except during Pass 1 and Pass 2, and it is empty again at the end of the episode.
- **The placemat:** flat in the middle of the tabletop, holding every item.
- **Waste bin:** on the tabletop against the right edge, within right arm reach.

Each arm owns the side it sits on. The right arm works the mat and the waste bin; the left arm works the tray and takes the glass. Items cross between them in the staging zone.

Vocabulary

Arm assignments

- **The glass is a left gripper item from start to finish.** The left gripper lifts it off the mat in Step 1 and loads it onto the tray in Step 7. The right gripper never touches it.
- **The right gripper owns the rest of the mat and the waste bin.** It lifts every other item off the mat, carries each dish to the waste bin to shake it, and sets each item down in its assigned spot in the staging zone.
- **The left gripper owns the tray.** It lifts every staged item and loads it onto the tray. Apart from the glass it never reaches the mat, and it never reaches the waste bin.
- **Cutlery and napkin:** the right gripper lifts the item off the mat and sets it down in the staging zone directly. No detour, no handoff.
- **Dishes:** the right gripper lifts the dish, carries it right to the waste bin, shakes it twice, then carries it left and sets it down in the dish lane. Still no handoff.
- **Placemat:** the only hand to hand transfer in the episode. The right gripper pinches a corner and lifts the bare mat clear of the table, passes it to the left arm above the staging zone with both grippers holding it for 1 second, and the left arm lays it on the tray. The transfer happens in the air, clear of the staged lanes. The mat never rests in the staging zone.
- An item is released only onto the tabletop, in the staging zone, or onto the tray. Never open a gripper over open space, and never over the waste bin.

Working a scattered mat

- **Type order, not position order.** Clear glass, then cutlery, then napkin, then dishes, no matter where each item happens to be lying. Do not clear by whatever is nearest.
- Within the cutlery, the order is always **fork, knife, spoon**. Within the dishes, always **bowl, small plate, large plate**.

- **Never pull an item out from under another.** If the item due next is pinned or covered, lift the covering item straight up first.
- **Napkin branch:** if the napkin is draped over the glass or over a utensil, take the napkin first, then return to the type order from the top. This is the only permitted departure from the order.
- Take each grip where the item is clear, not where the standard grip would be on a tidy table: a plate by any free stretch of rim, a utensil across any free part of its handle.
- Never drag an item across the mat to get a better grip. Lift it straight up.

Staging zone positions

Every item has a fixed spot in the staging zone. The zone is laid out in four lanes running from the tray side across to the mat side, and **the lanes are ordered by when the item leaves:** whatever is loaded onto the tray first sits nearest the tray. Nothing is ever lifted over another staged item.

- **Dish lane**, nearest the tray: the three dishes face up in a row, side by side and never stacked. Within the lane they read **large plate, small plate, bowl** outward from the tray, which is the order they will be loaded.
- **Cutlery lane**, next across: the three utensils flat and parallel, handle ends roughly in line, reading **fork, knife, spoon** outward from the tray.
- **Napkin spot**, next across: the crumpled napkin, clear of the cutlery lane.
- **Glass spot**, farthest from the tray and nearest the mat: the glass standing upright, clear of the napkin. It is the first item staged and the last one loaded, so it sits out of the traffic.

Leave a clear gap between lanes so each item can be lifted straight up without brushing its neighbors. Nothing in the staging zone touches, overlaps, or rests on anything else.

Tray zones

Once the placemat lines the tray, every item is lowered onto it in an assigned zone:

- **Dish stack:** centered on the mat, face up, largest at the bottom: large plate, then small plate, then bowl.
- **Cutlery:** laid flat together along the tray's near side, clear of the stack.
- **Glass:** standing upright in the tray's near corner, clear of the stack and the cutlery.
- **Napkin:** laid flat in the remaining corner beside the stack.

Nothing overhangs the tray's edge, and nothing rests on the bare tray outside the mat.

Handling standard

- Move exactly one item at a time.

- Carry every item level and lower it straight down. The only exception is a dish tilted face down over the waste bin to shake.
- Lower each item until it is resting on the mat or the stack, then release. Never drop an item onto the tray from above.
- Nothing is slid. Correct a bad placement by lifting the item and setting it down again.
- An item is settled when it stays still for 2 seconds after release.

Steps

Step 1: Clear the glass to staging

Goal: the glass stands upright at the glass spot and the tallest obstacle is off the mat.

- If the napkin is draped over the glass, take the napkin first (see Step 3), then come back here.
- The **left gripper** grips the glass around its body, lifts it straight up clear of whatever is around it, and carries it level to the **glass spot**, the end of the staging zone farthest from the tray.
- Lower it straight down and release it once it stands upright and still.
- This is the only time the left arm reaches the mat. It stays clear of the mat for the rest of the episode, and the right arm never touches the glass.

Check: the glass stands upright and stable in the glass spot, leaving the rest of the staging zone clear for the three lanes to come. If it rocks or leans, lift it and set it down again.

Step 2: Clear the cutlery to staging

Goal: the fork, knife, and soup spoon lie in the cutlery lane.

- Move one utensil at a time in this order, wherever each one happens to be lying on the mat: **fork, knife, spoon.**
- If the utensil due next is pinned under the napkin or another utensil, lift the covering item straight up and set it clear on the mat, then take the utensil in order.
- If a utensil is resting on a dish, lift it straight up off the dish. Do not slide it off.
- The **right gripper** grips each utensil across a free part of its handle, lifts it clear, and lowers it flat in the **cutlery lane**.
- Build the lane outward from the tray: the fork nearest the tray, the knife beside it, the spoon beyond that. The three lie flat and parallel with their handle ends roughly in line, clear of the glass. Do not stack them on each other.

Check: all three utensils lie flat in the cutlery lane, in fork, knife, spoon order outward from the tray, with a clear gap to the glass. Nothing was dragged across the mat to get to them.

Step 3: Clear the cloth napkin to staging

Goal: the napkin is off the mat.

- The **right gripper** pinches the napkin at a raised fold or a free edge, lifts it clear of the table, and lowers it at the **napkin spot**, between the cutlery lane and the glass.
- The napkin stays crumpled. Do not flatten or fold it here.
- If it comes up with a utensil caught in it, set the napkin back down, free the utensil, and lift the napkin again.

Check: no part of the napkin is left on the mat, and it sits in its own spot rather than draped over the cutlery or touching the glass.

Step 4: Shake and stage the three dishes

Goal: the bowl, small plate, and large plate have each been shaken over the waste bin and are sitting in the staging zone.

Work one dish at a time in this order, wherever each one is lying: **bowl, then small plate, then large plate**. The **right gripper** does the whole step; the left arm stays clear.

4.1 Lift and carry to the waste bin

- The **right gripper** grips the dish by any free stretch of rim and lifts it level off the mat.
- It carries the dish right, to over the waste bin.

4.2 Shake it over the waste bin

- Tilt the dish face down toward the bin.
- Give it **two short shakes**, then return the dish to level. Whatever comes off, comes off.

4.3 Carry it across to staging

- The **right gripper** returns the dish to level, carries it left across the bare mat, and lowers it face up in the **dish lane**, the lane nearest the tray.
- Fill the lane from the outside in, so each dish lands nearer the tray than the one before: the **bowl** at the outer end of the lane, the **small plate** beside it toward the tray, the **large plate** nearest the tray. That leaves the row reading large plate, small plate, bowl outward, which is the order they will be loaded.
- The three sit side by side and never stacked, with a clear gap to the cutlery lane.

Check: all three dishes are face up in the dish lane, separate, and reading large plate, small plate, bowl outward from the tray. Each was tilted and shaken over the waste bin on its way there. The mat is bare.

Expected state: the staging zone reads, from the tray side across: large plate, small plate, bowl in the dish lane; fork, knife, spoon in the cutlery lane; the crumpled napkin; the upright glass. The mat is bare and the tabletop around it is clear. The tray is still empty.

Step 5: Lay the placemat on the tray

Goal: the placemat lies flat on the tray as the base layer for everything else.

- The **right gripper** pinches one corner of the bare mat and lifts it clear of the table.
- It passes the mat to the **left arm** above the staging zone, clear of the staged lanes, both grippers holding it for 1 second before the right one releases.
- The **left gripper** carries the mat to the tray and lowers it flat, then releases.
- If the mat lands folded, bunched, or hanging over an edge, lift it by a corner and lay it down again.

Check: the mat lies flat on the tray with no fold standing up and no overhang. The middle of the tabletop is now completely bare.

Step 6: Load the dish stack

Goal: the three dishes are stacked face up on the mat, largest at the bottom.

- Load in this order, the reverse of the order they came off the table: **large plate, then small plate, then bowl**. That is the dish lane worked from the tray side outward, so no dish is lifted over another.
- The **left gripper** grips each dish by its rim, lifts it level out of the dish lane, and carries it to the tray.
- Lower each dish straight down until it rests on the mat or on the dish below it, then release.
- Center each dish on the one below it so the stack sits square rather than leaning.

Check: the stack is large plate, small plate, bowl, all face up and centered on the mat. If the stack leans or a dish rocks, lift that dish and lower it again. Do not shove it into place.

Step 7: Load the cutlery, napkin, and glass

Goal: the tray is fully loaded and the staging zone is empty.

- Load in this order: **cutlery, napkin, glass**.
- **Cutlery:** the **left gripper** lifts one utensil at a time from the cutlery lane, working from the tray side outward so the fork comes first, and lays each flat along the tray's near side, beside the stack rather than on it. The three lie together.
- **Napkin:** the **left gripper** pinches the napkin at its spot, lifts it, and lowers it into the remaining corner beside the stack.
- **Glass:** the **left gripper** grips the glass around its body at the glass spot, carries it upright, and lowers it into the tray's near corner until it rests on the mat. Release it only once it stands upright and still. The glass never goes on top of the stack.

Check: the staging zone is empty. Every item is on the tray, in its zone, resting on the mat or the stack. The glass is upright. Nothing overhangs the tray's edge.

Step 8: End the episode

Goal: the table is clear and both arms are safely home.

- Confirm the middle of the tabletop and the staging zone are both bare.
- Confirm the tray holds the mat, the three dish stack, the three utensils, the napkin, and the upright glass, and that the waste bin holds only shaken off scraps.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unload the tray. Put the placemat back flat in the middle of the tabletop.
2. Empty the waste bin if scraps are near the rim.
3. Scatter the tableware again on the mat: the three dishes loose and unstacked, the three utensils loose with at least one resting on a dish, and the napkin crumpled. Vary the scatter between episodes rather than rebuilding the same arrangement.
4. Put loose, dry scraps back to the three dishes.
5. Confirm every item sits fully on the mat with nothing on the bare tabletop.
6. Wipe the tabletop and clear the staging zone.
7. Clean or replace anything chipped, cracked, bent, torn, or stained.
8. Put the empty tray back against the left edge and the waste bin against the right edge.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Cleared by position instead of by type

- **Visible cue:** items come off the mat in whatever order they happen to be reachable (a dish taken before the cutlery, the napkin before the glass, cutlery split around a dish) rather than glass, then all cutlery, then napkin, then dishes.
- **SOP rule broken:** Steps 1 to 4, clear by type order: glass, cutlery, napkin, dishes, no matter where each item is lying, and never clear by whatever is nearest.
- **Coaching note:** the scatter changes every episode; the order never does.

Violation: Wrong order within a type

- **Visible cue:** the utensils leave in any order other than fork, knife, spoon, or the dishes leave in any order other than bowl, small plate, large plate.
- **SOP rule broken:** Steps 2 and 4, cutlery always fork, knife, spoon, and dishes always bowl, small plate, large plate.
- **Coaching note:** name the next item before you reach for it.

Violation: Item pulled out from under another

- **Visible cue:** a utensil is dragged out from beneath the napkin or another utensil, a dish is pulled out from under something resting on it, or an item is slid sideways to free it, instead of the covering item being lifted straight up first.
- **SOP rule broken:** Step 2, never pull an item out from under another, lift the covering item straight up first, then take the pinned item in type order.
- **Coaching note:** uncover it, then lift it. Never tug it free.

Violation: Napkin branch misused

- **Visible cue:** the napkin is taken out of order when it was not covering the glass or a utensil, or the napkin is left in place and the glass or a utensil is pulled out from under it.
- **SOP rule broken:** Steps 1 and 3, take the napkin first only when it is draped over the glass or a utensil, then return to the type order from the top.
- **Coaching note:** the napkin jumps the queue only when it is covering something.

Violation: Item staged in the wrong spot

- **Visible cue:** a staged item is not in its assigned lane or spot: a dish outside the dish lane or stacked on another dish, a utensil outside the cutlery lane or crossing another utensil, the napkin on the cutlery or against the glass, the glass anywhere but the far end. Also counts when a lane reads in the wrong order, such as the bowl nearest the tray or the spoon ahead of the fork.
- **SOP rule broken:** Steps 1 to 4, dish lane nearest the tray reading large plate, small plate, bowl outward, cutlery lane next reading fork, knife, spoon outward, napkin spot next, glass spot farthest from the tray, lanes ordered by when the item leaves, with a clear gap between them.

- **Coaching note:** put each item in its lane as it arrives; the staging zone is not a pile.

Violation: Staged item lifted over another

- **Visible cue:** during Steps 6 and 7 the left gripper reaches across or over a staged item to reach one behind it, most often taking the bowl or the spoon before the items nearer the tray.
- **SOP rule broken:** Steps 6 and 7, work each lane from the tray side outward, large plate, small plate, bowl, then fork, knife, spoon, then the napkin, then the glass, and nothing is lifted over another staged item.
- **Coaching note:** always take the nearest item in the lane; if you are reaching over something, the order is wrong.

Violation: Wrong arm used for an item

- **Visible cue:** the right gripper picks up or loads the glass; the left gripper lifts anything other than the glass off the mat, or reaches toward the waste bin; the right gripper reaches the tray or loads an item onto it.
- **SOP rule broken:** Steps 1 to 7, the glass is a left gripper item from start to finish, the right gripper owns the rest of the mat and the waste bin, and the left gripper owns the tray and otherwise stays off the mat.
- **Coaching note:** the glass is always the left arm's; everything else on the mat is the right arm's.

Violation: Unprescribed handoff

- **Visible cue:** the glass, a utensil, the napkin, or a dish is passed hand to hand instead of being set down in the staging zone by the right arm and picked up again by the left, or the placemat crosses without a handoff, or an arm releases before the 1 second hold is complete.
- **SOP rule broken:** Steps 1 to 7, only the placemat transfers hand to hand, with both grippers holding it for 1 second, and every other item is set down in the staging zone and picked up from there.
- **Coaching note:** the mat is the only item that changes hands in mid air.

Violation: Dish not shaken over the waste bin

- **Visible cue:** a dish goes from the mat to staging without being carried right to the waste bin and tilted face down, or it is tilted somewhere other than over the waste bin.
- **SOP rule broken:** Steps 4.1 and 4.2 (the right gripper carries each dish to the waste bin and tilts it face down before the dish is staged).
- **Coaching note:** every dish goes right before it goes left, including the bowl.

Violation: Scraps shaken outside the waste bin

- **Visible cue:** a dish is tilted and shaken while it is over the tabletop, the mat, the staging zone, or the tray, and scraps land outside the waste bin. Also counts when a dish is tilted on the carry back across to staging.
- **SOP rule broken:** Steps 4.1, 4.2, and 4.3 (tilt only while over the waste bin; return the dish to level before carrying it across).
- **Coaching note:** level it before you leave the bin.

Violation: Item put in the wrong place

- **Visible cue:** the napkin, a utensil, the glass, a dish, or the mat is released into the waste bin, or scraps are dumped onto the tray.
- **SOP rule broken:** Steps 4 to 7, the tray takes the mat, dishes, cutlery, napkin, and glass, and the waste bin takes only shaken off scraps and never receives an object.
- **Coaching note:** nothing but scraps ever enters the waste bin.

Violation: Wrong tray load order

- **Visible cue:** an item goes onto the tray before the placemat, or the dish stack is built in any order other than large plate, small plate, bowl, or the cutlery, napkin, and glass go on before the stack is complete.
- **SOP rule broken:** Steps 5, 6, and 7 (the placemat goes on first as the base layer; then the dish stack largest to smallest; then the cutlery, then the napkin, then the glass).
- **Coaching note:** mat first, then build the stack big to small, then the loose items.

Violation: Placemat not used as the tray base

- **Visible cue:** the mat is laid on the tray folded, bunched, or overhanging an edge and is left that way, or items are loaded onto the bare tray beside the mat instead of onto it.
- **SOP rule broken:** Step 5 and Tray zones (lay the mat flat on the tray, lift it and lay it again if it lands folded or overhanging, and lower every item onto the mat).
- **Coaching note:** flatten the mat before anything lands on it.

Violation: Placemat lifted before it is bare

- **Visible cue:** the mat is lifted, tugged, or shifted while any item is still on it, including using the mat to slide items toward an arm.
- **SOP rule broken:** Step 5, the right gripper lifts the mat only after Step 4 leaves it bare, and nothing is slid.
- **Coaching note:** the mat is the last thing off the table, and it comes off empty.

Violation: More than one item moved at once

- **Visible cue:** two utensils, two dishes, or a dish with a utensil still on it are lifted together, or the napkin comes up with something caught in it and is carried anyway.

- **SOP rule broken:** Steps 1 to 7, move exactly one item at a time, and if the napkin lifts a utensil with it, set it down, free the utensil, and lift again.
- **Coaching note:** one item per trip, every trip.

Violation: Item dropped onto the tray instead of lowered

- **Visible cue:** a gripper opens above the tray and the item falls the rest of the way, instead of being lowered until it rests on the mat or the stack.
- **SOP rule broken:** Steps 5 to 7, lower each item until it is resting, then release, and never drop an item onto the tray from above.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Dish stack built incorrectly

- **Visible cue:** the stack on the tray is out of size order, a dish is face down, a dish is clearly off center on the one below it, or the stack leans and is left that way.
- **SOP rule broken:** Step 6 and Tray zones (stack face up on the mat, large plate, small plate, bowl, each centered on the one below it, and lift any dish and lower it again that leaves the stack leaning).
- **Coaching note:** square each dish on the one under it before releasing.

Violation: Loose item stacked on the dishes

- **Visible cue:** a utensil, the napkin, or the glass ends up resting on top of the dish stack rather than in its own zone on the mat, or the glass is left lying on its side.
- **SOP rule broken:** Step 7 and Tray zones (cutlery flat along the near side, napkin in the remaining corner, glass upright in the near corner, all clear of the stack).
- **Coaching note:** the stack is not a shelf; every loose item gets its own zone.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, nudges, or drags an item across the mat, the tabletop, or the tray, either to reposition it for a better grip or to correct a placement, instead of lifting it clear and lowering it again. The item stays in contact with the surface while it moves.
- **SOP rule broken:** Steps 1 to 7, carry every item level and lower it straight down, never drag an item across the mat to get a grip, and correct a bad placement by lifting the item and setting it down again.
- **Coaching note:** lift it clear and put it down again; nothing gets shoved into position.

Violation: Item dropped, collided, or overlapped

- **Visible cue:** an item falls, strikes another object on the way, knocks the glass or a staged item over, or comes to rest overlapping something it should be clear of or overhanging the tray's edge.

- **SOP rule broken:** Steps 1 to 7, keep each item controlled on a clear travel path and release only after a stable placement clear of everything else.
- **Coaching note:** check the path and the landing spot before lowering.

Violation: Staging zone left occupied

- **Visible cue:** the episode reaches Step 8 with an item still sitting in the staging zone, or an item is left behind on the tabletop where the mat was.
- **SOP rule broken:** Step 7 and Step 8 (the staging zone is empty and the middle of the tabletop is bare before the episode ends).
- **Coaching note:** sweep the staging zone with your eyes before homing.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an item missing from the tray, an arm away from home, or no 3 second hold on the cleared table and loaded tray.
- **SOP rule broken:** Step 8 (confirm the bare tabletop and staging zone and the fully loaded tray; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective object:** a chipped, cracked, or bent item found mid episode, or a tray or waste bin that shifts off its zone. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Move the glass from the mat to staging
2. Move one utensil from the mat to staging
3. Uncover a pinned item by lifting what is on top of it
4. Move the cloth napkin from the mat to staging
5. Carry one dish from the mat to the waste bin
6. Shake one dish over the waste bin
7. Carry one shaken dish across to staging
8. Hand the placemat from the right arm to the left arm
9. Lay the placemat flat on the tray

10. Stack one dish on the tray
11. Load one utensil onto the tray
12. Load the napkin onto the tray
13. Load the glass upright onto the tray
14. Return both arms home and end the episode